

Antonio Bianchi

Curriculum Vitae

antoniob@purdue.edu
<https://antoniobianchi.me>
Last update: March 2026

Research Interests

My research interests span the domains of **Software and Systems Security**. In particular, I have been focusing on the broad area of **security of smart devices and edge devices**, such as smartphones, smartwatches, IoT devices, robotic vehicles (e.g., drones), and embedded systems. By employing formal verification, program analysis, and user studies, my work has provided a comprehensive approach to securing modern computing environments.

Education

- Ph.D., Thesis: “Identifying and Mitigating Trust Violations in the Mobile Ecosystem,” Computer Science Department, UC Santa Barbara, July 2018.
Advisors: Prof. Giovanni Vigna and Prof. Christopher Kruegel.
- M.S., Computer Science, University of Illinois at Chicago, May 2012.
- M.S., Computer Engineering, Politecnico di Milano, Italy, April 2012.
- B.Sc., Computer Engineering, Politecnico di Milano, Italy, September 2008.

Previous Positions

- ASST PROFESSOR, Department of Computer Sciences, Purdue University, West Lafayette, IN, August 2019 – July 2025.
- ASST PROFESSOR, Department of Computer Sciences, The University of Iowa, Iowa City, IA, August 2018 – July 2019.

Present Position

- ASSOC PROFESSOR, Department of Computer Sciences, Purdue University, West Lafayette, IN, August 2025 – present.

Awards and Honors

- Seed for Success Acorn Award for researchers having received a sponsored grant equal to or greater than \$1 million, from Purdue’s Office of Research. Grant: “FLARE: A Multi-layered Approach to Exploit-chain Generation (PI),” 2025.
- NSF, CNS Division of Computer and Network Systems Award, Faculty Early Career Development Program (CAREER), 2025.
- Qualified for the DARPA/ARPA-H AIXCC Final event as part of the Shellphish team (7 teams qualified out of 42 participating teams), 2024.
- Seed for Success Acorn Award for researchers having received a sponsored grant equal to or greater than \$1 million, from Purdue’s Office of Research. Grant: “FIREFLY: A Cyber-Physical Framework for Scalable CPS Modeling and Simulation (co-PI),” 2024.
- Undergraduate Advising Award, from Purdue CS Department, 2024.
- Seed for Success Acorn Award for researchers having received a sponsored grant equal to or greater than \$1 million, from Purdue’s Office of Research. Grant: “DICER: Directed Compilation for Assured Patching (PI),” 2023.
- Best Poster Award — MITRE 2023 embedded Capture the Flag (eCTF), 2023.
- Best Paper Award — USENIX Workshop on Offensive Technologies (WOOT), 2020.
- NSF, Directorate for Computer and Information Science and Engineering (CISE), Research Initiation Initiative (CRII) Award, 2019.
- Distinguished Paper Award — Network & Distributed System Security Symposium (NDSS), 2016.
- Third Place (First Place Self-funded Team) at the DARPA Cyber Grand Challenge, as part of the Shellphish team, 2016.
- Regents Special Fellowship — University of California, Santa Barbara, 2012.

Publications

Note: * indicates primary author, ^P denotes author who is postdoctoral researcher (or on other pre-faculty position) *at the time of writing*. Similarly, ^U and ^G indicate authors who were undergraduate and graduate students, respectively *at the time of writing*. Additionally, ^A denotes advisees and ^M denotes PhD advisors.

Articles Refereed In Conference Proceedings

1. Han Dai^{AG*}, Soumyakant Priyadarshan^P, Abdullah Imran^A, Ruoyu Wang, Antonio Bianchi. SCRIBE: Practical Static Binary Patching via Binary-Aware Recompilation of Decompiled Code. To appear in *IEEE European Symposium on Security and Privacy (Euro S&P)*, 2026. (acceptance rate TBD)
2. Zeyu Lei^{AG*}, Güiliz Seray Tuncay, Abdullah Imran^A, Z. Berkay Celik, Antonio Bianchi. A First Look at the Mobile Driving License (mDL) Standard and its Real-world Usage. To appear in *ACM Asia Conference on Computer and Communications Security (AsiaCCS)*, 2026. (acceptance rate TBD)
3. Greg Tystahl^{G*}, Jonah Ghebremichael^G, Siddharth Muralee^{AG}, Sourag Cherupattamoolayil^G, Antonio Bianchi, Aravind Kumar Machiry, Alexandros Kapravelos, William Enck. COSSETER: GitHub Actions Permission Reduction Using Demand-Driven Static Analysis. To appear in *USENIX Security Symposium (UsenixSEC)*, 2026. (acceptance rate TBD)
4. Georgios Androutsopoulos^{AG*}, Antonio Bianchi. deepSURF: Detecting Memory Safety Vulnerabilities in Rust Through Fuzzing LLM-Augmented Harnesses. To appear in *IEEE Symposium on Security and Privacy (IEEE S&P)*, 2026. (acceptance rate TBD)

5. Devin Ersoy^{G*}, Brandon Lee^{AG*}, Ananth Shreekumar^G, Arjun Arunasalam^G, Muhammad Ibrahim^{AG}, Antonio Bianchi, Z. Berkay Celik. Investigating the Impact of Dark Patterns on LLM-Based Web Agents. To appear in *IEEE Symposium on Security and Privacy (IEEE S&P)*, 2026. (acceptance rate TBD)
6. Saurabh R. Tanksalkar^{G*}, Siddharth Muralee^{AG}, Samuel Danduri^G, Praise Amusuo^G, Antonio Bianchi, James C. Davis, and Aravind Kumar Machiry. LEMIX: Enabling Testing of Embedded Applications as Linux Applications. In *USENIX Security Symposium (UsenixSEC)*, 2025. (17.1% acceptance rate)
7. Doguhan Yeke^{G*}, Kartik A. Pant^G, Muslum Ozgur Ozmen^G, Hyungsub Kim, James M. Goppert, Insup Hwang, Antonio Bianchi, and Z. Berkay Celik. Automated Discovery of Semantic Attacks in Multi-Robot Navigation Systems. In *USENIX Security Symposium (UsenixSEC)*, 2025. (17.1% acceptance rate)
8. Ruoyu Wu^{*A}, Muqi Zou^G, Arslan Khan, Taegyu Kim, Dongyan Xu, Dave (Jing) Tian, and Antonio Bianchi. NEUROSCOPE: Reverse Engineering Deep Neural Network on Edge Devices using Dynamic Analysis. In *USENIX Security Symposium (UsenixSEC)*, 2025. (17.1% acceptance rate)
9. Jianliang Wu^{*A}, Ruoyu Wu^A, Dongyan Xu, Dave (Jing) Tian, and Antonio Bianchi. Securing Bluetooth Across Layers: A Systematic Review of Protection Strategies. In *GetMobile: Mobile Computing and Communications*, 2025.
10. Haoyu Lu^{*G}, Hao Cai^G, Yichen Liang^G, Antonio Bianchi, and Z. Berkay Celik. A Progressive Transformer for Unifying Binary Code Embedding and Knowledge Transfer. In *IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)*, 2025. (25.5% acceptance rate)
11. Muslum Ozgur Ozmen^{*G}, Mehmet Oguz Sakaoglu^G, Jure Bizjak^G, Jianliang Wu^A, Antonio Bianchi, Dave (Jing) Tian, and Z. Berkay Celik. Why Am I Seeing Double? An Investigation of Device Management Flaws in Voice Assistant Platforms. In *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2025. (26% acceptance rate)
12. Hongwei Wu^{*AG}, Ruoyu Wu^{AG}, Jianling Wu^A, Ayushi Sharma^G, Aravind Machiry, and Antonio Bianchi. VeriBin: Adaptive Verification of Patches at the Binary Level. In *Network & Distributed System Security Symposium (NDSS)*, 2025. (16.1% acceptance rate)
13. Zeyu Lei^{*AG}, Güliz Seray Tuncay, Beatrice Carissa Williem^U, Z. Berkay Celik, and Antonio Bianchi. ScopeVerif: Analyzing the Security of Android's Scoped Storage via Differential Analysis. In *Network & Distributed System Security Symposium (NDSS)*, 2025. (16.1% acceptance rate)
14. Jianliang Wu^{*A}, Patrick Traynor, Dongyan Xu, Dave (Jing) Tian, and Antonio Bianchi. Finding Traceability Attacks in the Bluetooth Low Energy Specification and its Implementations. In *USENIX Security Symposium (UsenixSEC)*, 2024. (18.32% acceptance rate)
15. Reham Mohamed^{*G}, Arjun Arunasalam^G, Habiba Farrukh^G, Jason Tong, Antonio Bianchi, and Z. Berkay Celik. ATTention Please! An Investigation of the App Tracking Transparency Permission. In *USENIX Security Symposium (UsenixSEC)*, 2024. (18.32% acceptance rate)
16. Muqi Zou^{*G}, Arslan Khan^G, Ruoyu Wu^{AG}, Han Gao^G, Antonio Bianchi, and Dave (Jing) Tian. D-Helix: Improving Decompilation Accuracy via Symbolic Model Differentiation and Automatic Tuning. In *USENIX Security Symposium (UsenixSEC)*, 2024. (18.32% acceptance rate)
17. Abdullah Imran^{*AG} and Antonio Bianchi. Automated detection of cryptographic inconsistencies in Android's Keymaster implementations. In *Annual International Conference on Mobile Systems, Applications, and Services (MobiSys)*, 2024. (16.3% acceptance rate)
18. Hyungsub Kim^{*AG}, Rwitam Bandyopadhyay^G, Muslum Ozgur Ozmen^G, Z. Berkay Celik, Antonio Bianchi, Yongdae Kim, and Dongyan Xu. A Systematic Study of Physical Sensor Attack Hardness. In *IEEE Symposium on Security and Privacy (S&P)*, 2024. (17.8% acceptance rate)
19. Jianliang Wu^{*AG}, Ruoyu Wu^{AG}, Dongyan Xu, Dave (Jing) Tian, and Antonio Bianchi. SoK: The Long Journey of Exploiting and Defending the Legacy of King Harald Bluetooth. In *IEEE Symposium on Security and Privacy (S&P)*, 2024. (17.8% acceptance rate)

20. Doguhan Yeke^{*AG}, Muhammad Ibrahim^{AG}, Güliz Seray Tuncay, Habiba Farrukh^G, Abdullah Imran^{AG}, Antonio Bianchi, and Z. Berkay Celik. Wear's my Data? Understanding the Cross-Device Runtime Permission Model in Wearables. In *IEEE Symposium on Security and Privacy (S&P)*, 2024. (17.8% acceptance rate)
21. Prashast Srivastava^{*AG}, Flavio Toffalini, Kostyantyn Vorobyov, François Gauthier, Antonio Bianchi, and Mathias Payer. Crystallizer: A Hybrid Path Analysis Framework To Aid in Uncovering Deserialization Vulnerabilities. In *ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2023. (21% acceptance rate)
22. Hammas Bin Tanveer^{*G}, Mike Puchol, Rachee Singh, Antonio Bianchi, and Rishab Nithyanand. Making Sense of Constellations: Methodologies for Understanding Starlinks Scheduling Algorithms. In *Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2023.
23. Kyungtae Kim^{*G}, Sungwoo Kim^G, Kevin R. B. Butler, Antonio Bianchi, Rick Kennell, and Dave (Jing) Tian. Fuzz The Power: Dual-role State Guided Black-box Fuzzing for USB Power Delivery. In *USENIX Security Symposium (UsenixSEC)*, 2023. (29.2% acceptance rate)
24. Habiba Farrukh^{*G}, Reham Mohamed^G, Aniket Nare, Antonio Bianchi, and Z. Berkay Celik. LocIn: Inferring Semantic Location from Spatial Maps in Mixed Reality. In *USENIX Security Symposium (UsenixSEC)*, 2023. (29.2% acceptance rate)
25. Siddharth Muralee^{*AG}, Igibek Koishybayev^{*G}, Aleksandr Nahapetyan^G, Greg Tystahl^G, Brad Reaves, Antonio Bianchi, William Enck, Alexandros Kapravelos, and Aravind Machiry. ARGUS: A Framework for Staged Static Taint Analysis of GitHub Workflows and Actions. In *USENIX Security Symposium (UsenixSEC)*, 2023. (29.2% acceptance rate)
26. Ruoyu Song^{*AG}, Muslum Ozgur Ozmen^G, Hyungsub Kim^{AG}, Raymond Muller^G, Z. Berkay Celik, and Antonio Bianchi. Discovering Adversarial Driving Maneuvers against Autonomous Vehicles. In *USENIX Security Symposium (UsenixSEC)*, 2023. (29.2% acceptance rate)
27. Hyungsub Kim^{*AG}, Muslum Ozgur Ozmen^G, Z. Berkay Celik, Antonio Bianchi, and Dongyan Xu. PatchVerif: Discovering Faulty Patches in Robotic Vehicles. In *USENIX Security Symposium (UsenixSEC)*, 2023. (29.2% acceptance rate)
28. Arslan Khan^{*G}, Muqi Zou^G, Kyungtae Kim^G, Dongyan Xu, Antonio Bianchi, and Dave Jing Tian. Fuzzing SGX Enclaves via Host Program Mutations. In *IEEE European Symposium on Security and Privacy (EuroS&P)*, 2023. (35% acceptance rate)
29. Muhammad Ibrahim^{*AG}, Andrea Continella, and Antonio Bianchi. AoT - Attack on Things: A security analysis of IoT firmware updates. In *IEEE European Symposium on Security and Privacy (EuroS&P)*, 2023. (35% acceptance rate)
30. Priyanka Bose^{*G}, Dipanjan Das^G, Saastha Vasan^G, Sebastiano Mariani^G, Ilya Grishchenko^P, Andrea Continella, Antonio Bianchi, Christopher Kruegel^M, and Giovanni Vigna^M. COLUMBUS: Android App Testing Through Systematic Callback Exploration. In *International Conference on Software Engineering (ICSE)*, 2023, 2023. (26% acceptance rate)
31. Hyungsub Kim^{*AG}, Muslum Ozgur Ozmen^G, Antonio Bianchi, Z. Berkay Celik, and Dongyan Xu. Demo: Discovering Faulty Patches in Robotic Vehicle Control Software. In *Symposium on Vehicle Security and Privacy (VehicleSec)*, colocated with NDSS, 2023.
32. Muslum Ozgur Ozmen^{*G}, Habiba Farrukh^G, Hyungsub Kim^{AG}, Antonio Bianchi, and Z. Berkay Celik. Short: Rethinking Secure Pairing in Drone Swarms. In *Symposium on Vehicles Security and Privacy (VehicleSec)*, 2023.
33. Ruoyu Wu^{*AG}, Taegyu Kim^G, Dave (Jing) Tian, Antonio Bianchi, and Dongyan Xu. DnD: Decompiling Deep Neural Network Compiled Binary. Peer-reviewed Talk at *BlackHat Europe*, London, UK, 2022.
34. Prashast Srivastava^{*AG}, Stefan Nagy^G, Matthew Hicks, Antonio Bianchi, and Mathias Payer. One Fuzz Doesnt Fit All: Optimizing Directed Fuzzing via Target-tailored Program State Restriction. In *Annual Computer Security Applications Conference (ACSAC)*, 2022. (24.1% acceptance rate)

35. Ruoyu Wu^{*AG}, Taegyu Kim^G, Dave (Jing) Tian, Antonio Bianchi, and Dongyan Xu. DnD: A Cross-Architecture Deep Neural Network Decompiler. In *USENIX Security Symposium (UsenixSEC)*, 2022. (18.1% acceptance rate)
36. Abdullah Imran^{*AG}, Habiba Farrukh^G, Muhammad Ibrahim^{AG}, Z. Berkay Celik, and Antonio Bianchi. SARA: Secure Android Remote Authorization. In *USENIX Security Symposium (UsenixSEC)*, 2022. (18.1% acceptance rate)
37. Trung Nguyen^{*U}, Kyungtae Kim^G, Antonio Bianchi, and Dave (Jing) Tian. TruEMU: An Extensible, Open-Source, Whole-System iOS Emulator. Peer-reviewed Talk at *BlackHat*, Las Vegas, NV, 2022.
38. Sungwoo Kim^{*G}, Gisu Yeo^G, Taegyu Kim^G, Junghwan “John” Rhee^G, Yuseok Jeon, Antonio Bianchi, Dongyan Xu, and Dave (Jing) Tian. ShadowAuth: Backward-Compatible Automatic CAN Authentication for Legacy ECUs. In *Asia Conference on Computer and Communications Security (AsiaCCS)*, 2022. (18.3% acceptance rate)
39. Hyungsub Kim^{*AG}, Muslum Ozgur Ozmen^G, Z. Berkay Celik, Antonio Bianchi, and Dongyan Xu. PGPATCH: Policy-Guided Logic Bug Patching for Robotic Vehicles. In *IEEE Symposium on Security and Privacy (S&P)*, 2022. (14.5% acceptance rate)
40. Kyungtae Kim^{*G}, Ertza Warraich^G, Taegyu Kim^G, Byoungyoung Lee, Kevin Butler, Antonio Bianchi, and Dave (Jing) Tian. FUZZUSB: Hybrid Stateful Fuzzing of the Linux USB Gadget Stack. In *IEEE Symposium on Security and Privacy (S&P)*, 2022. (14.5% acceptance rate)
41. Jianliang Wu^{*AG}, Ruoyu Wu^{AG}, Dongyan Xu, Dave (Jing) Tian, and Antonio Bianchi. Formal Model-Driven Discovery of Bluetooth Protocol Design Vulnerabilities. In *IEEE Symposium on Security and Privacy (S&P)*, 2022. (14.5% acceptance rate)
42. Hyungsub Kim^{*AG}, Muslum Ozgur Ozmen^G, Antonio Bianchi, Z. Berkay Celik, and Dongyan Xu. Demo: Policy-based Discovery and Patching of Logic Bugs in Robotic Vehicles. In *Automotive and Autonomous Vehicle Security Workshop (AutoSec)*, colocated with NDSS, 2022.
43. Michael Reeves^{*G}, Dave (Jing) Tian, Antonio Bianchi, and Z. Berkay Celik. Towards Improving Container Security by Preventing Runtime Escapes. In *IEEE Secure Development Conference (SecDev)*, 2021.
44. Onur Zungur^{*G}, Antonio Bianchi, Gianluca Stringhini, and Manuel Egele. APPJITSU: Investigating the Resiliency of Android Applications. In *European IEEE Symposium on Security and Privacy (EuroS&P)*, 2021. (32% acceptance rate)
45. Jianliang Wu^{*AG}, Ruoyu Wu^{*AG}, Daniele Antonioli, Mathias Payer, Nils Ole Tippenhauer, Dongyan Xu, Dave (Jing) Tian, and Antonio Bianchi. LIGHTBLUE: Automatic Profile-Aware Debloating of Bluetooth Stacks. In *USENIX Security Symposium (UsenixSEC)*, 2021. (18.7% acceptance rate)
46. Arslan Khan^{*G}, Hyungsub Kim^{AG}, Byoungyoung Lee, Dongyan Xu, Antonio Bianchi, and Dave (Jing) Tian. M2MON: Building a MMIO-based Security Reference Monitor for Cyber-Physical Systems. In *USENIX Security Symposium (UsenixSEC)*, 2021. (18.7% acceptance rate)
47. Muhammad Ibrahim^{*AG}, Abdullah Imran^{AG}, and Antonio Bianchi. SafetyNOT: On the Usage of the SafetyNet Attestation API in Android. In *ACM International Conference on Mobile Systems, Applications, and Services (MobiSys)*, 2021. (21.7% acceptance rate)
48. Nilo Redini^{*G}, Andrea Continella, Aravind Machiry, Giulio De Pasquale^G, Dipanjan Das^G, Antonio Bianchi, Christopher Kruegel^M, and Giovanni Vigna^M. Diane: Identifying Fuzzing Triggers in Apps for Effective Vulnerability Analysis of IoT Devices. In *IEEE Symposium on Security and Privacy (S&P)*, 2021. (12.1% acceptance rate)
49. Hyungsub Kim^{*AG}, Muslum Ozgur Ozmen^G, Antonio Bianchi, Z. Berkay Celik, and Dongyan Xu. PGFUZZ: Policy-Guided Fuzzing for Robotic Vehicles. In *Network & Distributed System Security Symposium (NDSS)*, 2021. (15.2% acceptance rate)

50. Lei Zeyu^{*AG}, Yuhong Nan^P, Yanick Fratantonio, and Antonio Bianchi. On the Insecurity of SMS One-Time Password Messages against Local Attackers in Modern Mobile Devices. In *Network & Distributed System Security Symposium (NDSS)*, 2021. (15.2% acceptance rate)
51. Jianliang Wu^{*AG}, Yuhong Nan^P, Vireshwar Kumar^P, Dave (Jing) Tian, Antonio Bianchi, Mathias Payer, and Dongyan Xu. BLESAs: Spoofing Attacks against Reconnections in Bluetooth Low Energy. In *USENIX Workshop on Offensive Technologies (WOOT)*, 2020.
Best Paper Award.
52. Dario Nisi^{*G}, Antonio Bianchi, and Yanick Fratantonio. Exploring Syscall-Based Semantics Reconstruction of Android Applications. In *International Symposium on Research in Attacks, Intrusions and Defenses (RAID)*, 2019. (22% acceptance rate)

— **Pre-Purdue Hire¹:**

53. Moritz Eckert^{*G}, Antonio Bianchi, Ruoyu Wang^G, Yan Shoshitaishvili^G, Christopher Kruegel^M, and Giovanni Vigna^M. HeapHopper: Bringing Bounded Model Checking to Heap Implementation Security. In *USENIX Security Symposium (UsenixSEC)*, 2018. (19.1% acceptance rate)
54. Antonio Bianchi^{*}, Yanick Fratantonio^G, Aravind Machiry, Christopher Kruegel^M, Giovanni Vigna^M, Simon Pak Ho Chung, and Wenke Lee. Broken Fingers: On the Usage of the Fingerprint API in Android. In *Network & Distributed System Security Symposium (NDSS)*, 2018. (21.5% acceptance rate)
55. Antonio Bianchi^{*}, Eric Gustafson^G, Yanick Fratantonio^G, Christopher Kruegel^M, and Giovanni Vigna^M. Exploitation and Mitigation of Authentication Schemes Based on Device-Public Information. In *Annual Computer Security Applications Conference (ACSAC)*, 2017. (19.7% acceptance rate)
56. Nilo Redini^{*G}, Aravind Machiry, Dipanjan Das^G, Yanick Fratantonio^G, Antonio Bianchi, Eric Gustafson^G, Yan Shoshitaishvili^G, Christopher Kruegel^M, and Giovanni Vigna^M. BootStomp: On the Security of Bootloaders in Mobile Devices. In *USENIX Security Symposium (UsenixSEC)*, 2017. (16.3% acceptance rate)
57. Aravind Machiry^{*}, Eric Gustafson^G, Chad Spensky^G, Chris Salls^G, Nick Stephens^U, Ruoyu Wang^G, Antonio Bianchi, Yung Ryn Choe, Christopher Kruegel^M, and Giovanni Vigna^M. BOOMERANG: Exploiting the Semantic Gap in Trusted Execution Environments. In *Network & Distributed System Security Symposium (NDSS)*, 2017. (16% acceptance rate)
58. Ruoyu Wang^{*G}, Yan Shoshitaishvili^G, Antonio Bianchi, Aravind Machiry, John Grosen^U, Paul Grosen^U, Christopher Kruegel^M, and Giovanni Vigna^M. Ramblr: Making Reassembly Great Again. In *Network & Distributed System Security Symposium (NDSS)*, 2017. (16% acceptance rate)
Distinguished Paper Award.
59. Yanick Fratantonio^{*G}, Antonio Bianchi, William Robertson, Engin Kirda, Christopher Kruegel^M, and Giovanni Vigna^M. TriggerScope: Towards Detecting Logic Bombs in Android Apps. In *IEEE Symposium on Security and Privacy (S&P)*, 2016. (13.3% acceptance rate)
60. Vitor Afonso^{*G}, Antonio Bianchi, Yanick Fratantonio^G, Adam Doupé^G, Mario Polino^G, Paulo de Geus, Christopher Kruegel^M, and Giovanni Vigna^M. Going Native: Using a Large-Scale Analysis of Android Apps to Create a Practical Native-Code Sandboxing Policy. In *Network & Distributed System Security Symposium (NDSS)*, 2016. (15.4% acceptance rate)
61. Simone Mutti^{*G}, Yanick Fratantonio^G, Antonio Bianchi, Luca Invernizzi^G, Jacopo Corbetta^G, Dhilung Kirat^G, Christopher Kruegel^M, and Giovanni Vigna^M. BareDroid: Large-Scale Analysis of Android Apps on Real Devices. In *Annual Computer Security Applications Conference (ACSAC)*, 2015. (24.4% acceptance rate)
62. Antonio Bianchi^{*}, Yanick Fratantonio^G, Christopher Kruegel^M, and Giovanni Vigna^M. NJAS: Sandboxing Unmodified Applications in non-rooted Devices Running Stock Android. In *ACM Workshop on Security and Privacy in Smartphones and Mobile Devices (SPSM)*, 2015.

¹I joined Purdue in August 2019.

63. Yanick Fratantonio^{*G}, Aravind Machiry, Antonio Bianchi, Christopher Kruegel^M, and Giovanni Vigna^M. CLAPP: Characterizing Loops in Android Applications. In *Symposium on the Foundations of Software Engineering (FSE)*, 2015. (25.4% acceptance rate)
64. Yanick Fratantonio^{*G}, Aravind Machiry, Antonio Bianchi, Christopher Kruegel^M, and Giovanni Vigna^M. CLAPP: Characterizing Loops in Android Applications. In *International Workshop on Software Development Lifecycle for Mobile (DeMobile)*, 2015.
65. Yanick Fratantonio^{*G}, Antonio Bianchi, William Robertson, Manuel Egele, Christopher Kruegel^M, Engin Kirda, and Giovanni Vigna^M. On the Security and Engineering Implications of Finer-Grained Access Controls for Android Developers and Users. In *Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA)*, 2015. (22.7% acceptance rate)
66. Antonio Bianchi^{*}, Jacopo Corbetta^G, Luca Invernizzi^G, Yanick Fratantonio^G, Christopher Kruegel^M, and Giovanni Vigna^M. What the App is That? Deception and Countermeasures in the Android User Interface. In *IEEE Symposium on Security and Privacy (S&P)*, 2015. (13.5% acceptance rate)
67. Yinzhi Cao^{*G}, Yanick Fratantonio^G, Antonio Bianchi, Manuel Egele, Christopher Kruegel^M, Giovanni Vigna^M, and Yan Chen. EdgeMiner: Automatically Detecting Implicit Control Flow Transitions through the Android Framework. In *Network & Distributed System Security Symposium (NDSS)*, 2015. (16.9% acceptance rate)
68. Sebastian Poeplau^{*U}, Yanick Fratantonio^G, Antonio Bianchi, Christopher Kruegel^M, and Giovanni Vigna^M. Execute This! Analyzing Unsafe and Malicious Dynamic Code Loading in Android Applications. In *Network & Distributed System Security Symposium (NDSS)*, 2014. (18.6% acceptance rate)
69. Antonio Bianchi^{*}, Yan Shoshitaishvili^G, Christopher Kruegel^M, and Giovanni Vigna^M. Blacksheep: Detecting Compromised Hosts in Homogeneous Crowds. In *ACM Conference on Computer and Communications Security (CCS)*, 2012. (18.9% acceptance rate)

Preprints (not published elsewhere)

1. Muqi Zou^{G*}, Hongyu Cai^G, Hongwei Wu^{AG}, Zion Leonahenahe Basque^G, Arslan Khan, Z. Berkay Celik, Dave (Jing) Tian, Antonio Bianchi, Ruoyu Wang, Dongyan Xu. D-LiFT: Improving LLM-based Decompiler Backend via Code Quality-driven Fine-tuning. *arXiv preprint arXiv:2506.10125*, 2025.
2. Muhammad Ibrahim^{AG*}, Güliz Seray Tuncay, Z. Berkay Celik, Aravind Kumar Machiry, Antonio Bianchi. LM-Scout: Analyzing the Security of Language Model Integration in Android Apps. *arXiv preprint arXiv:2505.08204*, 2025.
3. Siddharth Muralee^{AG*}, Sourag Cherupattamoolayil^G, James C. Davis, Antonio Bianchi, Aravind Kumar Machiry. Reactive Bottom-Up Testing. *arXiv preprint arXiv:2509.03711*, 2025.
4. Ruoyu Song^{AG*}, Muslum Ozgur Ozmen^G, Hyungsub Kim, Antonio Bianchi, Z. Berkay Celik. Enhancing LLM-based Autonomous Driving Agents to Mitigate Perception Attacks. *arXiv preprint arXiv:2409.14488*, 2024.
5. Hongyu Cai^{G*}, Arjun Arunasalam^G, Leo Y. Lin^G, Antonio Bianchi, Z. Berkay Celik. Rethinking How to Evaluate Language Model Jailbreak. *arXiv preprint arXiv:2404.06407*, 2024.

Books, Book Chapters, Technical Reports, Theses

1. Yan Shoshitaishvili^{*G}, Antonio Bianchi^G, Kevin Borgolte^G, Amat Cama^G, Jacopo Corbetta^G, Francesco Disperati^G, Audrey Dutcher^G, John Grosen^U, Paul Grosen^U, Aravind Machiry, Chris Salls^G, Nick Stephens^U, Ruoyu Wang^G, and Giovanni Vigna^M. Mechanical Phish: Resilient Autonomous Hacking. In *IEEE Security & Privacy Magazine – SPSI: Hacking without Humans*, 2018.

2. Antonio Bianchi^G, Kevin Borgolte^G, Jacopo Corbetta^G, Francesco Disperati^G, Andrew Dutcher^G, John Grosen^U, Paul Grosen^U, Aravind Machiry, Christopher Salls^G, Yan Shoshitaishvili^G, Nick Stephens^U, Giovanni Vigna^M, and Ruoyu Wang^G. Cyber Grand Shellphish. In *Phrack Magazine*, 2017.

Invited Lectures

National and International Meetings

1. “Fuzzing SGX Enclaves via Host Program Mutations,” Invited Talk – Intel, Apr 2024.
2. “Security Analysis of Three Emerging Pieces of Android OS,” Invited Talk – Google, Oct 2022.
3. “From the analysis of mobile apps to the analysis of the mobile ecosystem,” Keynote Speaker – International Workshop on Security in Mobile Technologies at ACNS, Feb 2022.
4. “How not to use text messages for authentication,” Invited Talk – Android Security and Privacy Research (ASPIRE) Summit, Google, Mountain View, CA, Feb 2020.
5. “Securing Interconnected Software: from Mobile Apps to IoT Devices,” Invited Talk – Symantec, Dec 2019.
6. “Cyber Grand Shellphish: Shellphish and the DARPA CGC,” Peer-reviewed Talk - DEF CON, Las Vegas, NV, USA, Aug 2016.
7. “A Dozen Years of Shellphish – From DEFCON to the Cyber Grand Challenge,” Invited Talk - Nuit Du Hack, Paris, France, Jul 2016.
8. “A Dozen Years of Shellphish – From DEFCON to the Cyber Grand Challenge,” Peer-reviewed Talk - Chaos Communication Congress (CCC), Berlin, Germany, Dec 2015.
9. “A Dozen Years of Shellphish – From DEFCON to the Cyber Grand Challenge,” Invited Talk - HIT-CON Enterprise, Taipei, Taiwan, Aug 2015.

Universities and Other Institutions

1. “Research at PurSec Lab.” Invited Talk – Rose-Hulman Institute of Technology, Terre Haute, IN, Nov 2023.
2. “Machines Hacking Machines: Who Needs People?” Invited Talk – Rose-Hulman Institute of Technology, Terre Haute, IN, Oct 2019.
3. “Detecting Vulnerable Code: from Mobile Apps to IoT Devices.” Invited Talk – Grinnell College, Grinnell, IA, Sep 2018.

Other Professional Activities

1. **Organization of Security Competitions (CTFs).** Throughout my career, I have been involved in the organization of numerous security competitions, typically called Capture the Flag Competitions (CTFs, in short).
 - (a) DEF CON CTF (in 2019, 2020, 2021, 2024, and 2025): The largest CTF competition in the world with more than 500 teams participating online each year in the Qualification Event and 16 teams participating in-person to the Final Event.
 - (b) ACSAC CTF, 2024.
 - (c) UCSB iCTF (in 2024): Organized by the ACTION NSF AI Institute, the Shellphish CTF team, and the UCSB Women in Computer Science group. This competition focused on high-school and undergraduate participants.
 - (d) b01lers CTF (in 2022 and 2023): The Capture the Flag competitions organized by b01lers, the Purdue CTF team.
2. **Qualified to the Final Event of the DARPA/ARPA-H AI Cyber Challenge (AIxCC).**

Funding

1. Agency/Title of Grant:	NSF: CAREER: Toward a Principled Methodology for Trusted Software Patching
2. Duration of Funding:	04/01/2025 - 03/31/2030
3. Total Amount of Award:	\$580,000
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	DARPA – Intelligent Generation of Tools for Security (INGOTS): FLARE: A Multi-layered Approach to Exploit-chain Generation
2. Duration of Funding:	02/01/2025 - 01/31/2028
3. Total Amount of Award:	\$2,198,887
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	DARPA – Artificial Intelligence Cyber Challenge (AIxCC): Gift
2. Duration of Funding:	27/12/2024 - open ended
3. Total Amount of Award:	\$100,000
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	United States Military Academy West Point: Investigating Private Data Collection on Edge Devices and Its Impact on the Military Population
2. Duration of Funding:	09/01/2024 - 08/31/2026
3. Total Amount of Award:	\$117,532
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	Lockheed Martin Corp.: An End-to-end Pipeline for Lifting and Patching DNN Binaries for Adversarial and Defensive Applications
2. Duration of Funding:	04/1/2024 - 12/15/2024
3. Total Amount of Award:	\$121,000
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	DARPA – Artificial Intelligence Cyber Challenge (AIxCC): Gift
2. Duration of Funding:	03/15/2024 - open ended
3. Total Amount of Award:	\$50,000
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	DARPA – Faithful Integrated Reverse-Engineering and Exploitation (FIRE): FIREFLY: A Cyber-Physical Framework for Scalable CPS Modeling and Simulation
2. Duration of Funding:	11/30/2023 - 05/29/2027
3. Total Amount of Award:	\$6,500,087
4. Your Role:	Co-PI; award made to Purdue

1. Agency/Title of Grant:	Lockheed Martin Corp.: Towards an End-to-end Pipeline for Lifting and Patching DNN Binaries for Adversarial and Defensive Applications
2. Duration of Funding:	10/01/2023 - 12/22/2023
3. Total Amount of Award:	\$65,000
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	DOE: Enabling Secure and Resilient XFC: A Software/Hardware-security Co-design Approach
2. Duration of Funding:	06/01/2023 - 12/31/2023
3. Total Amount of Award:	\$80,000
4. Your Role:	Co-PI; subcontract from Virginia Tech
1. Agency/Title of Grant:	Google: Gift: Android Security and PrIvacy REsearch (ASPIRE) Award, Exploring the Implementation and Usage of Android Storage APIs
2. Duration of Funding:	10/15/2023 - open ended
3. Total Amount of Award:	\$90,000
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	ONR: Semantic Decompilation of Deep Neural Network Binaries and Its Adversarial and Defensive Implications
2. Duration of Funding:	01/01/2023 - 12/31/2025
3. Total Amount of Award:	\$750,655
4. Your Role:	Co-PI; award made to Purdue
1. Agency/Title of Grant:	Google: Gift: Android Security and PrIvacy REsearch (ASPIRE) Award, Improving the Security and Usability of the Wear OS Permission Model
2. Duration of Funding:	10/15/2022 - open ended
3. Total Amount of Award:	\$80,850
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	ONR: An Integrated Toolkit for IoT Protocol Dialecting with Formal Verification
2. Duration of Funding:	08/01/2022 - 07/31/2023
3. Total Amount of Award:	\$620,000
4. Your Role:	Co-PI; award made to Purdue
1. Agency/Title of Grant:	Google: Gift: Android Security and PrIvacy REsearch (ASPIRE) Award, Improving the Usability of Android APIs for Conformity of Standard Security Practices
2. Duration of Funding:	10/15/2021 - open ended
3. Total Amount of Award:	\$100,000
4. Your Role:	PI; award made to Purdue

1. Agency/Title of Grant:	DARPA – Assured Micropatching (AMP): DICER: Directed Compilation for Assured Patching
2. Duration of Funding:	08/05/2020 - 08/04/2024
3. Total Amount of Award:	\$3,869,685
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	Google: Gift: Google Security Rewards Program
2. Duration of Funding:	04/15/2020 - open ended
3. Total Amount of Award:	\$8,000
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	ONR: Bringing Fuzzing to the Cyber-Physical World
2. Duration of Funding:	01/15/2020 - 01/14/2023
3. Total Amount of Award:	\$799,877
4. Your Role:	Co-PI; award made to Purdue
1. Agency/Title of Grant:	DARPA – Computers and Humans Exploring Software Security (CHESS): CHECRS: Cognitive Human Enhancements for Cyber Reasoning Systems
2. Duration of Funding:	09/01/2019 - 03/31/2023
3. Total Amount of Award:	\$705,103
4. Your Role:	Co-PI; subcontract from Arizona State University
1. Agency/Title of Grant:	NSF: CRII Award: SaTC: Vetting and Improving the Usage of Trusted Execution Environments for Authentication in Mobile Devices
2. Duration of Funding:	06/01/2019 - 05/31/2021
3. Total Amount of Award:	\$174,972
4. Your Role:	PI; award made to Purdue
1. Agency/Title of Grant:	DARPA – Computers and Humans Exploring Software Security (CHESS): CHECRS: Cognitive Human Enhancements for Cyber Reasoning Systems
2. Duration of Funding:	11/29/2018 - 08/31/2019
3. Total Amount of Award:	\$175,838
4. Your Role:	Co-PI; subcontract from Arizona State University

Funding/Fellowships Received for Student Support.

1. Agency/Title of Grant:	Bloomberg: Infrastructure & Security Fellowship
2. Duration of Funding:	September 2025 – May 2026
3. Total Amount of Award:	\$42,000 and Full Tuition Coverage
4. Your Role:	PhD Student Advisor (Siddharth Muralee)
1. Agency/Title of Grant:	Gerondelis Foundation Scholarship
2. Duration of Funding:	Spring 2024
3. Total Amount of Award:	\$5,000
4. Your Role:	PhD Student Advisor (Georgios Androutsopoulos)
1. Agency/Title of Grant:	Caterpillar, Cyber + X Project with CERIAS: Intrusion Detection and Prevention Through Firmware Analysis
2. Duration of Funding:	Spring 2024 – Fall 2025
3. Total Amount of Award:	\$20,000
4. Your Role:	PhD Student Advisor (Han Dai)

Students

1 Graduated MS and PhD Students

1. Ashwin Nambiar, MS Graduated Fall 2025.
Last Known Position: Lucid Motors (Security Researcher)
2. Abdullah Imran, PhD Graduated Summer 2025.
PhD Thesis Title: Usability and Security of Android Trusted Execution Environments
Last Known Position: Teaching Assistant Professor at Purdue University
3. Zeyu Lei, PhD Graduated Summer 2025.
PhD Thesis Title: Evaluating Modern Mobile Security: A Multifaceted Approach Using Formal Methods and Empirical Analysis
Last Known Position: Palo Alto Networks
4. Trivikram Thirukkonda, MS Graduated Spring 2025.
MS. Thesis Title: FuzzGauge – A Method to Automatically Determine Reasons for a Fuzz Blocker
Last Known Position: Marvell Semiconductors (Firmware Engineer)
5. Muhammad Ibrahim, PhD Graduated Fall 2024.
PhD Thesis Title: Analyzing Secure and Attested Communication in Mobile Devices
Last Known Position: PostDoc at Georgia Institute of Technology (supervised by Prof. Brendan Saltaformaggio)
6. Ruoyu Wu, PhD Graduated Summer 2024 (co-advised with Prof. Dongyan Xu).
PhD Thesis Title: Towards Reverse Engineering Deep Neural Networks on Edge Devices
Last Known Position: Google (Software Engineer)
7. Hyungsub Kim, PhD Graduated Fall 2023 (co-advised with Prof. Dongyan Xu).
PhD Thesis Title: Defeating Cyber and Physical Attacks in Robotic Vehicles
Last Known Position: Tenure-track Assistant Professor at IU Bloomington
8. Jianliang Wu, PhD Graduated Summer 2023 (co-advised with Prof. Dongyan Xu).
PhD Thesis Title: Securing IoT Systems via Protocol Formal Analysis and Debloating
Last Known Position: Tenure-track Assistant Professor at Simon Fraser University

9. Prashast Srivastava, PhD Graduated Spring 2023 (co-advised with Prof. Mathias Payer).
PhD Thesis Title: Practical Methods for Dynamic Software Analysis of Real-world Systems
Last Known Position: PostDoc at Columbia University (supervised by Prof. Suman Jana)
10. Rowan Brock Hart, MS Graduated Fall 2022.
MS. Thesis Title: Fuzzing Deeper Logic with Impeding Function Transformation
Last Known Position: Intel (Security Engineer)

2 Undergraduate Research Supervision

1. Andrew Witt (CS) — Fall 2025
2. Neil Hommes (CS) — Fall 2025
3. William Boulton (CS) — Fall 2025
4. Jack Roscoe (CS) — Spring 2025
5. Michael Li (CS) — Spring 2025
6. Aditya Gupta (CS) — Spring 2025
7. Sooha Park (CS) — Spring 2025
8. Sravya Dandibhatta (CS) — Spring 2025
9. Leo Gu (CS) — Spring 2025
10. Aaryan Shah (CS) — Spring 2025
11. Trent Seaman (ECE) — Fall 2024
12. Nicholas Andry (CS) — Fall 2024
13. Philip Frey (CS) — Fall 2024
14. Glen Cheng (ECE) — Fall 2024
15. Burhanuddin Rajgara (CS) — Summer 2024
16. Bo-Shiun Yen (CIT) — Spring 2024
17. Derek Freeman (CS) — Summer 2023
18. Nithyashree Rangaprasad (CS) — Summer 2023
19. Everett Louis Johnson (CS) — Summer 2022
20. Zheng Shirong (CS) — Spring 2022
21. Beatrice Carissa Williem (CS) — Spring 2022
22. Dirk Jonathan (Jack) Locascio (CS) — Fall 2021
23. Trung Hoang Nguyen (CIT/CS) — Fall 2021
24. Rowan Brock Hart (CS) — Spring 2021
25. Xinwen Li (CS) — Summer 2020
26. Rowan Brock Hart (CS) — Summer 2020
27. Han Dai (CS) — Summer 2020
28. Alex Lin (CS) — Fall 2020
29. Connor McMillin (CS) — Spring 2020

3 Current and Previous Postdoc Associates

1. Soumyakant Priyadarshan, PhD from Stony Brook University, August 2024 – July 2025.
Topic: Binary Analysis and Binary Patching

4 Service on MS/PhD Committees

1. Zhong, Zheng (PhD Prelim), April 2026.
2. Zou, Muqi (PhD Final), February 2026.
3. Shen, Zilin (PhD Prelim), November 2025.
4. Song, Ruoyu (PhD Prelim), October 2025. [Co-Chair]
5. Imran, Abdullah (PhD Final), May 2025. [Chair]
6. Lei, Zeyu (PhD Final), June 2025. [Chair]
7. Fan, Yongming (PhD Prelim), April 2025.
8. Zou, Muqi (PhD Prelim), March 2025.
9. Ibrahim, Muhammad (PhD Final), September 2024. [Chair]
10. Ozmen, Muslum Ozgur (PhD Final), June 2024.
11. Mohamed, Reham (PhD Final), June 2024.
12. Wu, Ruoyu (PhD Final), June 2024. [Co-chair]
13. Wu, Xialong (PhD Prelim), April 2024.
14. Imran, Abdullah (PhD Prelim), April 2024. [Chair]
15. Lei, Zeyu (PhD Prelim), March 2024. [Chair]
16. Khan, Arslan (PhD Final), November 2023.
17. Kim, Hyungsub (PhD Final), November 2023. [Co-chair]
18. Mohamed, Reham (PhD Prelim), June 2024.
19. Alsharif, Khaled Serag (PhD Final), July 2023.
20. Ibrahim, Muhammad (PhD Prelim), July 2023. [Chair]
21. Farrukh, Habiba (PhD Final), June 2023.
22. Srivastava, Prashast (PhD Final), June 2023. [Co-chair]
23. Wu, Jianliang (PhD Final), May 2023. [Co-chair]
24. Ozmen, Muslum Ozgur (PhD Prelim), April 2023.
25. Kumar, Parvin (MS), April 2023.
26. Kim, Jiwon (PhD Prelim), April 2023.
27. Ye, Yapeng (PhD Prelim), December 2022.
28. Hart, Rowan Brock (MS), November 2022. [Chair]
29. Kim, Kyungtae (PhD Final), October 2022.
30. Kim, Hyungsub (PhD Prelim), October 2022. [Co-chair]
31. Wu, Ruoyu (PhD Prelim), July 2022. [Co-chair]
32. Khan, Arslan (PhD Prelim), April 2022.
33. Alsharif, Khaled Serag (PhD Prelim), March 2022.
34. Possemato, Andrea (PhD Final, at EURECOM, France), September 2021.
35. Kim, Kyungtae (PhD Prelim), September 2021.
36. Wu, Jianliang (PhD Prelim), April 2021. [Co-chair]
37. Cheng, Yicheng (PhD Final, at IUPUI), July 2021.
38. Reeves, Michael (MS), May 2021.

Teaching

Table 1 summarizes my primary teaching assignments.

In addition, I have taught or co-instructed the following “Special Topics” courses (CS59000 / CS49000):

- “LLMs for Software Security” – Spring 2025 – 5 undergraduate students (supporting the CS/Horizon3.ai collaboration).
- “Binary Hardening” – Spring 2025 – 3 undergraduate students (supporting the CERIAS/Caterpillar collaboration).
- “eCTF” (Co-instructor) – Spring 2023 – 25 CS/ECE students (supporting Purdue’s participation in the MITRE Embedded Capture the Flag competition).

Semester & Year	Course Number	Title of Course	Number of Students	Student Classification
<i>At Purdue:</i>				
Spring 2026	CS52700	Software Security	12	Graduate
Spring 2025	CS52700	Software Security	27	Graduate
Fall 2024	CS39700	Honors Seminar	55	Undergraduate
Fall 2024	CS42600	Computer Security	61	Undergraduate
Spring 2024	CS52700	Software Security	29	Graduate
Fall 2023	CS39700	Honors Seminar	31	Undergraduate
Fall 2023	CS49000-SWS	Software Security	10	Undergraduate
Spring 2023	CS52700	Software Security	29	Graduate
Fall 2022	CS39700	Honors Seminar	19	Undergraduate
Fall 2022	CS49000-SWS	Software Security	13	Undergraduate
Spring 2022	CS52700	Software Security	32	Graduate
Fall 2021	CS39700	Honors Seminar	33	Undergraduate
Fall 2021	CS59200-AST	Automated Security Testing	14	Graduate
Spring 2021	CS52700	Software Security	29	Graduate
Fall 2020	CS59100-SEC	CERIAS Security Seminar	15	Graduate
Spring 2020	CS52700	Software Security	21	Graduate
Fall 2019	CS59000-MSS	Mobile Systems and Smartphone Security	10	Graduate
<i>At the University of Iowa:</i>				
Spring 2019	CS4980	Mobile Systems and Smartphone Security	21	Graduate
Fall 2018	CS3620	Operating Systems	53	Undergraduate

Table 1: Teaching assignments.

CS39700 is a zero-credit course, CS59100-SEC is a 1 credit course, all other courses are 3 credit hours.

Service

1 University

- Faculty Advisor of “B01lers” (Purdue Capture the Flag Student Organization), 2019 – present.
- Instructor responsible for CS591-SEC “CERIAS Security Seminar”, 2020.

2 Professional

- Program Committee Co-Chair for Workshop on Offensive Technologies (WOOT): 2026.

- Program Committee for European Conference on Computer Systems (EuroSys): 2026.
- Invited Panelist: IEEE Secure Development Conference (SecDev), 2025.
- Student Advisor for the Speed Advising Session at the Midwest Security Workshop (MSW): 2019, 2024, 2025.
- NSF SAtC Panelist: 2024 and 2025.
- Program Committee Chair: Workshop on Binary Analysis Research (BAR) at the Network & Distributed System Security Symposium (NDSS), 2020.
- Program Committee Co-Chair: Workshop on Binary Analysis Research (BAR) at the Network & Distributed System Security Symposium (NDSS), 2019.
- Session Chair: NDSS 2024, VehicleSec 2024.
- Program Committee for IEEE Symposium on Security and Privacy (S&P): 2020, 2021, 2022, 2024, 2025.
- Program Committee for USENIX Security Symposium (UsenixSEC): 2020, 2021, 2024, 2025.
- Program Committee for Network & Distributed System Security Symposium (NDSS): 2020, 2021, 2023.
- Program Committee for ACM Conference on Computer and Communications Security (CCS): 2020, 2023.
- Program Committee for the Annual Computer Security Applications Conference (ACSAC): 2024.
- Program Committee for International Symposium on Research in Attacks, Intrusions and Defenses (RAID): 2023, 2024, 2025, and 2026.
- Program Committee for ACM ASIA Conference on Computer and Communications Security (AsiaCCS): 2021.
- Program Committee for European Symposium on Research in Computer Security (ESORICS): 2020, 2023.
- Program Committee for ACM Conference on Data and Application Security and Privacy (CODASPY): 2024.
- Program Committee for Symposium on Vehicle Security and Privacy (VehicleSec): 2023, 2024, 2025.
- Program Committee for European Workshop on Systems Security (EuroSec): 2023, 2024.
- Program Committee for Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA): 2022.
- Program Committee for Workshop on Offensive Technologies (WOOT): 2022, 2024.
- Program Committee for Workshop on the Internet of Safe Things: 2022.
- Program Committee for International Workshop on Security in Mobile Technologies (SecMT): 2020.
- Program Committee for Workshop on Binary Analysis Research (BAR) at NDSS: 2019, 2020, 2021, 2022, 2023.
- Paper Shepherding: S&P 2026, S&P 2025, VehicleSec 2025, ACSAC 2024, RAID 2024, ACSAC 2023, RAID 2023, USENIX 2022, NDSS 2022, S&P 2019.
- Journal Reviewer for ACM Transactions on Privacy and Security (TOPS): 2025, 2026.
- Journal Reviewer for IEEE Transactions on Information Forensics & Security (TIFS): 2025, 2026.
- Journal Reviewer for Computer & Security: 2024, 2025, 2026.
- Journal Reviewer for ACM Computing Surveys Review (CSUR): 2020, 2023.
- Journal Reviewer for IEEE Transactions on Knowledge and Data Engineering (TKDE): 2021.
- Journal Reviewer for IEEE Transactions on Mobile Computing (TMC): 2020.
- Journal Reviewer for IEEE Transactions on Dependable and Secure Computing (TDSC): 2018.